

1. Throughout this question, the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

- (a) For each of the following ARMA processes $\{X_t : t \in \mathbb{Z}\}$: identify the order of the process, write its equation using the backward shift operator, B , and determine whether or not the process is invertible. You should justify your answers and show your working clearly.

(i) $X_t = \frac{1}{2}X_{t-1} - \frac{1}{4}X_{t-2} + Z_t + \frac{1}{3}Z_{t-1}$

SOLUTION:

This is an $ARMA(2, 1)$ process.

Using the backward shift operator, the process is written

$$\left(1 - \frac{B}{2} + \frac{B^2}{4}\right) X_t = \left(1 + \frac{B}{3}\right) Z_t.$$

The process is invertible if the roots of

$$1 + \frac{z}{3} = 0$$

lie outside the unit circle. Since the root of the above equation is $z = -3$ (which lies outside the unit circle), the process is invertible.

(ii) $X_t = \frac{1}{5}X_{t-1} - \frac{3}{8}X_{t-2} + Z_t + \frac{7}{6}Z_{t-1} - \frac{1}{2}Z_{t-2}$

SOLUTION:

This is an $ARMA(2, 2)$ process.

Using the backward shift operator, the process is written

$$\left(1 - \frac{B}{5} + \frac{3B^2}{8}\right) X_t = \left(1 + \frac{7B}{6} - \frac{B^2}{2}\right) Z_t.$$

The process is invertible if the roots of

$$1 + \frac{7z}{6} - \frac{z^2}{2} = 0$$

lie outside the unit circle.

$$\begin{aligned} 1 + \frac{7z}{6} - \frac{z^2}{2} &= 0 \\ \implies 6 + 7z - 3z^2 &= 0 \\ \implies (3 - z)(2 + 3z) &= 0 \end{aligned}$$

Therefore the roots are $z = 3$ and $z = -2/3$. Since $-2/3$ lies inside the unit circle the process is not invertible.

TURN OVER

(b) The (weakly) stationary process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = -0.4Y_{t-1} + 0.32Y_{t-2} + Z_t.$$

(i) Write this process in the form

$$Y_t = \sum_{k=0}^{\infty} \psi_k Z_{t-k}$$

where the coefficients $\{\psi_k : k = 0, 1, 2, \dots\}$ are defined in terms of k . You should show your working clearly.

SOLUTION:

Write the process as

$$\begin{aligned} (1 + 0.4B - 0.32B^2) Y_t &= Z_t \\ \implies (1 - 0.4B)(1 + 0.8B) Y_t &= Z_t \end{aligned}$$

Then

$$\begin{aligned} Y_t &= \frac{1}{(1 - 0.4B)(1 + 0.8B)} Z_t \\ &= \left(\frac{A_1}{1 - 0.4B} + \frac{A_2}{1 + 0.8B} \right) Z_t \quad (\text{for some constants } A_1, A_2). \end{aligned}$$

Hence

$$\begin{aligned} A_1(1 + 0.8B) + A_2(1 - 0.4B) &= 1; \\ \implies A_1 + A_2 + (0.8A_1 - 0.4A_2)B &= 1. \end{aligned}$$

Equating coefficients we see that

$$\begin{aligned} A_1 + A_2 &= 1; \\ 0.8A_1 - 0.4A_2 &= 0. \end{aligned}$$

Then $A_2 = 1 - A_1$ and

$$\begin{aligned} 0.8A_1 - 0.4(1 - A_1) &= 0 \\ \implies 1.2A_1 &= 0.4 \\ \implies A_1 &= \frac{1}{3}. \end{aligned}$$

Therefore $A_2 = \frac{2}{3}$ and

$$Y_t = \frac{1}{3} \left[\frac{1}{1 - 0.4B} + \frac{2}{1 + 0.8B} \right] Z_t.$$

Let

$$f(B) = (1 + aB)^{-1} \quad (\text{for some constant } a)$$

CONTINUED

$$\begin{aligned}\implies f'(B) &= -a(1 + aB)^{-2} \\ \implies f''(B) &= 2!(-1)^2 a^2 (1 + aB)^{-3}\end{aligned}$$

In general, the k^{th} derivative of $f(B)$ is

$$f^{(k)}(B) = k!(-a)^k (1 + aB)^{-(k+1)}.$$

and the Taylor expansion of $f(B)$ about 0 is

$$\begin{aligned}f(B) &= f(0) + f'(0)B + \frac{1}{2!}f''(0)B^2 + \dots \\ &= 1 + \sum_{k=1}^{\infty} (-a)^k B^k.\end{aligned}$$

Hence

$$\begin{aligned}Y_t &= \frac{1}{3} \left[\frac{1}{1 - 0.4B} + \frac{2}{1 + 0.8B} \right] Z_t \\ &= \frac{1}{3} \left[1 + \sum_{k=1}^{\infty} 0.4^k B^k + 2 \left(1 + \sum_{k=1}^{\infty} (-0.8)^k B^k \right) \right] Z_t \\ &= Z_t + \frac{1}{3} \sum_{k=1}^{\infty} (0.4^k + 2(-0.8)^k) Z_{t-k}.\end{aligned}$$

- (ii) Using your answer to (i), determine the autocovariance function of $\{Y_t : t \in \mathbb{Z}\}$.

SOLUTION:

Suppose $h \geq 0$, then

$$\begin{aligned} \text{Cov}(Y_t, Y_{t+h}) &= \mathbb{E}(Y_t Y_{t+h}) - \mathbb{E}(Y_t) \mathbb{E}(Y_{t+h}) \\ &= \mathbb{E}(Y_t Y_{t+h}) \\ &= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} \psi_j Z_{t-j} \right) \left(\sum_{k=h}^{\infty} \psi_k Z_{t+h-k} \right) \right] \\ &= \sum_{j=0}^{\infty} \sum_{k=h}^{\infty} \psi_j \psi_k \mathbb{E}(Z_{t-j} Z_{t+h-k}). \end{aligned}$$

Since $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process, note that $\mathbb{E}(Z_{t-j} Z_{t+h-k}) = 0$ except where $j = k - h \implies k = j + h$.

Hence, the above sum can be written

$$\begin{aligned} \text{Cov}(Y_t, Y_{t+h}) &= \sum_{j=0}^{\infty} \psi_j \psi_{j+h} \mathbb{E}(Z_{t-j}^2) \\ &= \sigma_z^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+h}. \end{aligned}$$

Consider $\sum_{j=0}^{\infty} \psi_j \psi_{j+h}$. When $h = 0$:

$$\begin{aligned} \sum_{j=0}^{\infty} \psi_j \psi_{j+h} &= \sum_{j=0}^{\infty} \psi_j^2 \\ &= 1 + \frac{1}{9} \sum_{j=1}^{\infty} (0.4^j + 2(-0.8)^j)^2 \\ &= 1 + \frac{1}{9} \left[\sum_{j=1}^{\infty} (0.16)^j + 4 \sum_{j=1}^{\infty} (-0.32)^j + 4 \sum_{j=1}^{\infty} (0.64)^j \right] \\ &= 1 + \frac{1}{9} \left[\frac{0.16}{1 - 0.16} - \frac{4 \times 0.32}{1 + 0.32} + \frac{4 \times 0.64}{1 - 0.64} \right] \\ &= 1 + \frac{1}{9} \left[\frac{4}{21} - \frac{4 \times 8}{33} + \frac{4 \times 16}{9} \right] \\ &= 1 + \frac{4}{9} \left[\frac{33}{693} - \frac{168}{693} + \frac{1232}{693} \right] \\ &= 1 + \left(\frac{4}{9} \right) \left(\frac{1097}{693} \right) \\ &= 1 + \frac{4388}{6237} \\ &= \frac{10625}{6237}. \end{aligned}$$

CONTINUED

When $h > 0$:

$$\begin{aligned}
 \sum_{j=0}^{\infty} \psi_j \psi_{j+h} &= \psi_h + \frac{1}{9} \sum_{j=1}^{\infty} (0.4^j + 2(-0.8)^j) (0.4^{j+h} + 2(-0.8)^{j+h}) \\
 &= \frac{1}{3} (0.4^h + 2(-0.8)^h) + \frac{0.4^h}{9} \sum_{j=1}^{\infty} 0.16^j \\
 &\quad + \frac{2(0.4^h + (-0.8)^h)}{9} \sum_{j=1}^{\infty} (-0.32)^j + \frac{4(-0.8)^h}{9} \sum_{j=1}^{\infty} 0.64^j \\
 &= \frac{1}{3} (0.4^h + 2(-0.8)^h) + \frac{0.4^h}{9} \left[\frac{0.16}{1 - 0.16} \right] \\
 &\quad + \frac{2(0.4^h + (-0.8)^h)}{9} \left[\frac{(-0.32)}{1 + 0.32} \right] + \frac{4(-0.8)^h}{9} \left[\frac{0.64}{1 - 0.64} \right] \\
 &= \frac{1}{3} (0.4^h + 2(-0.8)^h) + \frac{0.4^h}{9} \left(\frac{4}{21} \right) + \frac{2(0.4^h + (-0.8)^h)}{9} \left(\frac{-8}{33} \right) \\
 &\quad + \frac{4(-0.8)^h}{9} \left(\frac{16}{9} \right) \\
 &= (0.4)^h \left(\frac{1}{3} + \frac{4}{189} - \frac{16}{297} \right) + (-0.8)^h \left(\frac{2}{3} - \frac{16}{297} + \frac{64}{81} \right) \\
 &= \left(\frac{625}{2079} \right) (0.4^h) + \left(\frac{1250}{891} \right) (-0.8)^h.
 \end{aligned}$$

By symmetry (since the process is (weakly) stationary), for $h < 0$

$$\sum_{j=0}^{\infty} \psi_j \psi_{j+h} = \left(\frac{625}{2079} \right) (0.4^{|h|}) + \left(\frac{1250}{891} \right) (-0.8)^{|h|}.$$

Hence the autocovariance function is

$$\gamma(h) = \begin{cases} \sigma_z^2 \frac{10625}{6237} & h = 0; \\ \sigma_z^2 \left[\left(\frac{625}{2079} \right) (0.4)^{|h|} + \left(\frac{1250}{891} \right) (-0.8)^{|h|} \right] & h = \pm 1, \pm 2, \dots \end{cases}$$

2. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

(a) The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = \frac{1}{3}X_{t-1} + Z_t + \frac{8}{3}Z_{t-1} - Z_{t-2}$$

- (i) Determine the spectral density function of $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION:

The process can be written

$$\begin{aligned} \left(1 - \frac{B}{3}\right) X_t &= \left(1 + \frac{8B}{3} - B^2\right) Z_t \\ \phi(B)X_t &= \theta(B)Z_t \\ \implies X_t &= \frac{\theta(B)}{\phi(B)}Z_t \end{aligned}$$

where

$$\phi(B) = 1 - \frac{B}{3}; \quad \theta(B) = 1 + \frac{8B}{3} - B^2.$$

The spectral density function is given by

$$g_X(w) = \frac{\sigma_z^2}{2\pi} \frac{\theta(e^{iw})}{\phi(e^{iw})} \frac{\theta(e^{-iw})}{\phi(e^{-iw})}$$

where $w \in [-\pi, \pi]$. Then

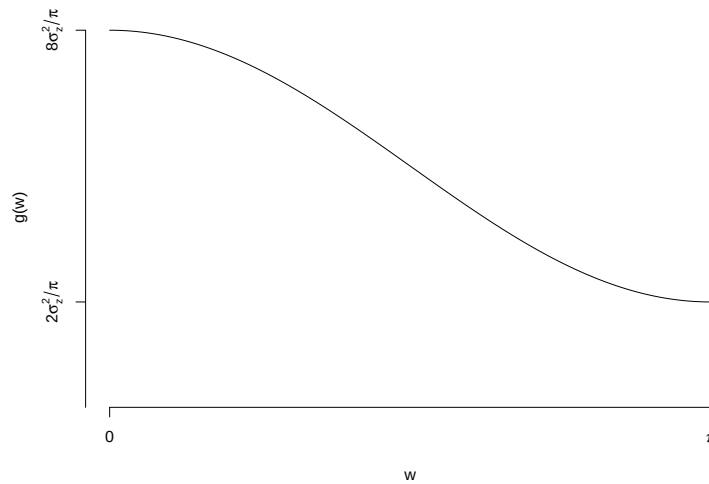
$$\begin{aligned} g_X(w) &= \frac{\sigma_z^2}{2\pi} \frac{\left(1 + \frac{8e^{iw}}{3} - e^{2iw}\right) \left(1 + \frac{8e^{-iw}}{3} - e^{-2iw}\right)}{\left(1 - \frac{e^{iw}}{3}\right) \left(1 - \frac{e^{-iw}}{3}\right)} \\ &= \frac{\sigma_z^2}{2\pi} \frac{\left(2 + \frac{64}{9} + \frac{8}{3}(e^{iw} + e^{-iw}) - \frac{8}{3}(e^{iw} + e^{-iw}) - (e^{i2w} + e^{-i2w})\right)}{\frac{10}{9} - \frac{1}{3}(e^{iw} + e^{-iw})} \\ &= \frac{\sigma_z^2}{2\pi} \left[\frac{\frac{82}{9} - 2\cos(2w)}{\frac{10}{9} - \frac{2}{3}\cos w} \right] \\ &= \frac{\sigma_z^2}{2\pi} \left[\frac{\frac{82}{9} - 2(2\cos^2 w - 1)}{\frac{10}{9} - \frac{2}{3}\cos w} \right] \\ &= \frac{\sigma_z^2}{2\pi} \left[\frac{\frac{100}{9} - 4\cos^2 w}{\frac{10}{9} - \frac{2}{3}\cos w} \right] \\ &= \frac{\sigma_z^2}{2\pi} \left[\frac{\left(\frac{10}{3} - 2\cos w\right) \left(\frac{10}{3} + 2\cos w\right)}{\frac{1}{3} \left(\frac{10}{3} - 2\cos w\right)} \right] \\ &= \frac{\sigma_z^2}{2\pi} (10 + 6\cos w). \end{aligned}$$

CONTINUED

- (ii) Sketch the spectral density function from (i) for frequency values, w , where $w \in [0, \pi]$.

SOLUTION:

A sketch of the spectral density function is



- (iii) Using the spectral density function, determine the variance of the process $\{X_t : t \in \mathbb{Z}\}$ in terms of σ_z^2 .

SOLUTION:

The variance of the process is given by

$$\text{Var}(X_t) = \int_{-\pi}^{\pi} g(w) dw.$$

Then

$$\begin{aligned} \text{Var}(X_t) &= \frac{\sigma_z^2}{2\pi} \int_{-\pi}^{\pi} (10 + 6 \cos w) dw \\ &= \frac{\sigma_z^2}{2\pi} [10w + 6 \sin w]_{-\pi}^{\pi} \\ &= \frac{\sigma_z^2}{2\pi} (10\pi + 10\pi) \\ &= 10\sigma_z^2. \end{aligned}$$

- (b) A linear filter is applied to the process $\{X_t : t \in \mathbb{Z}\}$ from (a) where

$$W_t = \frac{1}{2}X_t - \frac{1}{2}X_{t-1}.$$

- (i) Determine the spectral density function of the process $\{W_t : t \in \mathbb{Z}\}$.

TURN OVER

SOLUTION:

We write the process as

$$W_t = \sum_{j=-\infty}^{\infty} b_j X_{t-j}$$

and observe that $b_0 = \frac{1}{2}$, $b_1 = -\frac{1}{2}$ and $b_j = 0$ otherwise.

Then, the transfer function is

$$\begin{aligned} h(w) &= \sum_{j=-\infty}^{\infty} b_j e^{-iwj} \\ &= \frac{1}{2} - \frac{1}{2} e^{-iw}. \end{aligned}$$

The squared gain is

$$\begin{aligned} |h(w)|^2 &= h(w) \overline{h(w)} \\ &= \left(\frac{1}{2} - \frac{1}{2} e^{-iw} \right) \left(\frac{1}{2} - \frac{1}{2} e^{iw} \right) \\ &= \frac{1}{4} + \frac{1}{4} - \frac{1}{4} (e^{iw} + e^{-iw}) \\ &= \frac{1}{2} (1 - \cos w). \end{aligned}$$

The spectral density function of $\{W_t\}$ is given by

$$\begin{aligned} g_W(w) &= |h(w)|^2 g_X(w) \\ &= \frac{1}{2} (1 - \cos w) \frac{\sigma_z^2}{2\pi} (10 + 6 \cos w) \\ &= \frac{\sigma_z^2}{4\pi} (1 - \cos w) (10 + 6 \cos w) \end{aligned}$$

for $w \in (-\pi, \pi)$

- (ii) Briefly describe the effect of the linear filter on the original process $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION:

Since $1 - \cos w = 0$ at $w = 0$ and $1 - \cos w$ is maximised ($=2$) when $w = \pm\pi$, it follows that the linear filter will remove or dampen low frequency behaviour and amplify high frequency behaviour.

3. Throughout this question, the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .
- (a) The process $\{X_t\}$, where t is a calendar year, measures the annual number of transactions (in tens of thousands) made on a company's online shopping website and is an ARMA(1,1) process of the form

$$X_t = \mu + \phi(X_{t-1} - \mu) + Z_t + \theta Z_{t-1}.$$

Here, μ , ϕ and θ are real-valued parameters.

Data on past annual numbers of transactions are collected and a model of the above form is fitted, with parameter estimates: $\hat{\mu} = 20.1$, $\hat{\phi} = 0.6$, $\hat{\theta} = 0.4$ and $\hat{\sigma}_z^2 = 1.1$. The following table shows annual numbers of transactions for the years 2018–2022:

Year	2018	2019	2020	2021	2022
No. of transactions (tens of thousands)	20.1	19.6	20.5	20.3	19.8

Using the above information, provide an optimal forecast for the number of transactions in 2023. Here, the term ‘optimal’ implies that the forecasted value is that which minimises the mean square prediction error at a given time point. You should show your working clearly.

SOLUTION:

We can define $Y_t = X_t - \mu$ and re-write the process as

$$Y_t - \phi Y_{t-1} = Z_t + \theta Z_{t-1}.$$

We define $[Z_{n+k}]$ to be

$$[Z_{n+k}] = \begin{cases} Z_{n+k} & k \leq 0; \\ 0 & k > 0 \end{cases}$$

where $k \in \mathbb{Z}$.

Then, if $\hat{Y}_{n+k}(n)$ denotes the predicted value of Y_t at $t = n + k$, having observed the series to time n , we know that

$$\hat{Y}_{n+k} - \phi \hat{Y}_{n+k-1} = [Z_{n+k}] + \theta [Z_{n+k-1}].$$

For $k = 1$ (year 2023) it follows that

$$\begin{aligned} \hat{Y}_{2023} &= \phi \hat{Y}_{2022} + \theta Z_{2022} \\ &= \phi Y_{2022} + \theta Z_{2022} \quad (\text{since } Y_{2022} \text{ is observed}). \end{aligned}$$

Consider Z_t : since the process can be written

$$(1 - \phi B)Y_t = (1 + \theta B)Z_t$$

TURN OVER

it follows that

$$Z_t = \frac{(1 - \phi B)}{(1 + \theta B)} Y_t.$$

Let $f(B) = (1 + \theta B)^{-1}$, then

$$\begin{aligned} f'(B) &= -\theta(1 + \theta B)^{-2} \\ f''(B) &= 2\theta^2(1 + \theta B)^{-3} \\ f'''(B) &= (-1)3!\theta^3(1 + \theta B)^{-4}. \end{aligned}$$

In general, the k^{th} derivative of $f(B)$ is

$$f^{(k)}(B) = (-1)^k k! \theta^k (1 + \theta B)^{-(k+1)}.$$

Then Taylor series expansion of $f(B)$ about 0 yields

$$\begin{aligned} f(B) &= f(0) + f^{(1)}(0)B + \frac{1}{2!}f^{(2)}(0)B^2 + \frac{1}{3!}f^{(3)}(0)B^3 + \dots \\ &= 1 - \theta B + \theta^2 B^2 - \theta^3 B^3 + \dots \\ &= \sum_{j=0}^{\infty} (-\theta)^j B^j. \end{aligned}$$

Then

$$\begin{aligned} Z_t &= \frac{(1 - \phi B)}{1 + \theta B} Y_t \\ &= (1 - \phi B) \sum_{j=0}^{\infty} (-\theta)^j B^j Y_t \\ &= \sum_{j=0}^{\infty} (-\theta)^j B^j Y_t + \sum_{j=0}^{\infty} (-1)^{j+1} \phi \theta^j B^{j+1} Y_t \\ &= Y_t + \sum_{j=1}^{\infty} [(-\theta)^j + (-1)^j \phi \theta^{j-1}] Y_{t-j} \\ &= Y_t + \sum_{j=1}^{\infty} [(-1)^j (\theta^j + \phi \theta^{j-1})] Y_{t-j}. \end{aligned}$$

Hence

$$\begin{aligned} \hat{Y}_{2023} &= \phi Y_{2022} + \theta \left[Y_{2022} + \sum_{j=1}^{\infty} [(-1)^j (\theta^j + \phi \theta^{j-1})] Y_{2022-j} \right] \\ &= (\theta + \phi) Y_{2022} + \sum_{j=1}^{\infty} [(-1)^j (\theta^{j+1} + \phi \theta^j)] Y_{2022-j}. \end{aligned}$$

Using the estimate $\hat{\mu} = 20.1$ and the data provided, a table of values of $Y_t = X_t - \hat{\mu}$ is Using these data (and the estimates $\hat{\phi} = 0.6$ and $\hat{\theta} = 0.4$) we deduce that

$$\begin{aligned} \hat{Y}_{2023} &= (0.4 + 0.6)Y_{2022} - (0.4^2 + 0.6(0.4))Y_{2021} + (0.4^3 + 0.6(0.4)^2)Y_{2020} \\ &\quad - (0.4^4 + 0.6(0.4^3))Y_{2019} \end{aligned}$$

CONTINUED

Year	2018	2019	2020	2021	2022
Y_t	0	-0.5	0.4	0.2	-0.3

$$\begin{aligned}
&= Y_{2022} - 0.4Y_{2021} + 0.16Y_{2020} - 0.064Y_{2019} \\
&= -0.3 - (0.4 \times 0.2) + (0.16 \times 0.4) + (0.064 \times 0.5) \\
&= -0.284.
\end{aligned}$$

Then

$$\begin{aligned}
\hat{X}_{2023} &= -0.284 + \hat{\mu} \\
&= -0.284 + 20.1 \\
&= 19.816 \approx 19.8.
\end{aligned}$$

Therefore the predicted number of transactions in 2023 is 19800.

(b) The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = \lambda Y_{t-3} + Z_t$$

where λ is a real-valued parameter.

(i) Determine the set of values of λ for which this process is (weakly) stationary. Justify your answer.

SOLUTION:

The process is weakly stationary if the roots of

$$1 - \lambda z^3 = 0$$

lie outside the unit circle.

Hence

$$\begin{aligned}
&\lambda z^3 = 1 \\
\implies z^3 &= \frac{1}{\lambda} \\
\implies z &= |\lambda|^{-1/3} e^{i \frac{2k\pi}{3}} \quad k = 0, 1, 2
\end{aligned}$$

and the roots lie outside the unit circle when

$$\begin{aligned}
&\frac{1}{\lambda^{1/3}} > 1 \quad \text{or} \quad \frac{1}{\lambda^{1/3}} < -1 \\
\implies -1 &< \lambda^{1/3} < 1 \\
\implies -1 &< \lambda < 1.
\end{aligned}$$

(ii) Assuming that the process $\{Y_t : t \in \mathbb{Z}\}$ is (weakly) stationary, with appropriate justification determine the partial autocorrelation function (PACF) for this

TURN OVER

process. You may use, without proof, that the autocovariance of $\{Y_t : t \in \mathbb{Z}\}$ at lag h ($h \in \mathbb{Z}$), denoted $\gamma_Y(h)$, is

$$\gamma_Y(h) = \begin{cases} \frac{\sigma_z^2 \lambda^{\frac{|h|}{3}}}{1-\lambda^2} & h = 3k \text{ (where } k \in \mathbb{Z}) \\ 0 & \text{otherwise.} \end{cases}$$

SOLUTION:

Let $\psi(h)$ denote the PACF at lag h . Examining the autocovariance function, the autocovariance at lags 1 and 2 is zero and it follows that the PACF will also be zero at lags 1 and 2.

At lag 3, given that the autocovariance is zero at lags 1 and 2, it follows that the PACF at lag 3 is equal to the ACF at lag 3. Then

$$\psi(3) = \frac{\gamma_Y(3)}{\gamma_Y(0)} = \lambda.$$

(Alternatively, one can argue that the PACF at lag 3 is the coefficient of Y_{t-3} in the autoregressive linear model

$$Y_t = \lambda Y_{t-3} + Z_t$$

At lags > 3 , the PACF will be zero. This is because the PACF at lag m is the coefficient of Y_{t-m} in the autoregressive model

$$Y_t = \beta_{m,1}Y_{t-1} + \beta_{m,2}Y_{t-2} + \dots + \beta_{m,m}Y_{t-m} + Z_t$$

since we have $Y_t = \lambda Y_{t-3} + Z_t$ it follows that $\beta_{m,m}$ will be zero for all $m > 3$.

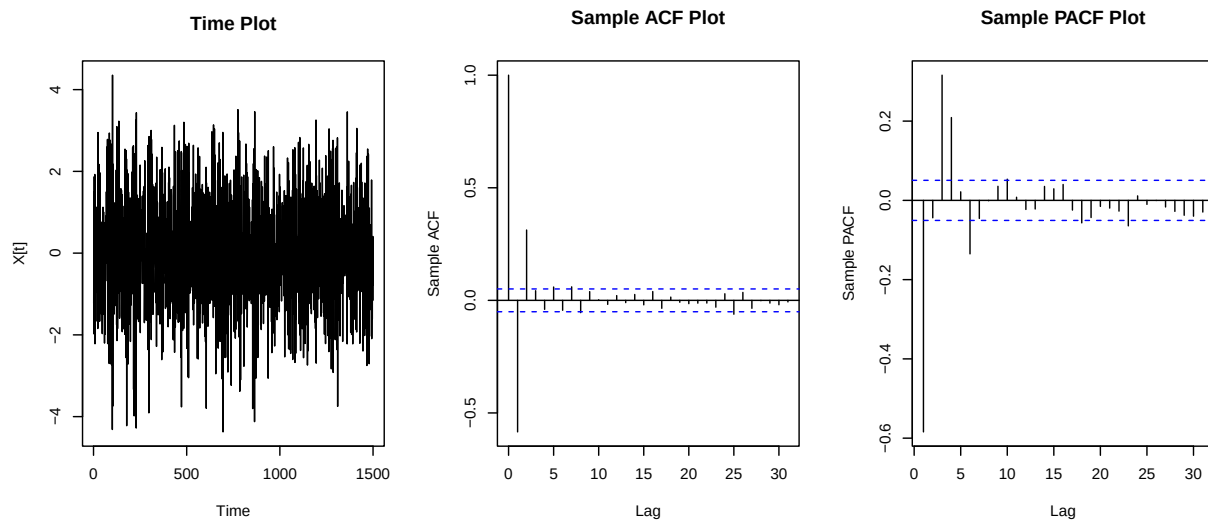
- (iii) Suppose that a large sequential sample of data from this process is collected where $\lambda = 0.7$. Using these data the sample PACF is calculated for lags from 1 to 15 and a plot of the sample PACF against the lag is produced. Describe how you would expect this plot to look. Justify your answer.

SOLUTION:

The sample PACF should be close to zero at all lags not equal to 3. At lag 3, there should be a spike with a sample PACF value of approximately 0.7.

4. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

- (a) A set of 1500 sequentially sampled values of the time series $\{X_t : t \in \mathbb{Z}\}$ is obtained. The figure below shows a time plot, a plot of the sample autocorrelation function (ACF) against the lag and a plot of the sample partial autocorrelation function (PACF) against the lag, for the sampled values.



- (i) Using the plots above, suggest an appropriate model for the time series. You should justify your answer and write out the equation of your model using suitable parameters but you do not need to provide parameter estimates.

SOLUTION:

The time plot suggests that the series is (weakly) stationary and does not contain seasonal terms, so we'd want to fit a stationary model.

Looking the ACF plot, the ACF appears to cut to zero after lag 2. This suggests that there is an MA component of order 2 in the model (and probably not an AR component).

Examining the PACF plot, the PACF does not cut off to zero immediately, which further suggests that there is an MA component to the model.

Taking all of the above into account, we might suggest that an $MA(2)$ model is fitted to these data of the form

$$X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}.$$

- (ii) Suppose that you fit the model suggested in (i) to the sampled data. Describe how you would check that the model is a good fit to the data. Note: you do not need to refer to R commands or other computer code to answer this question.

TURN OVER

SOLUTION:

The model residuals, $\hat{Z}_t$, should be calculated (where $\hat{Z}_t = [\hat{\theta}(B)]^{-1}X_t$) and a time plot of the residuals should be produced. We'd expect this time plot to look like that for a white noise process.

If the model fits well, the residuals should be uncorrelated so we would produce a plot of the sample ACFs at different lags for the residuals and these sample ACFs should be close to zero.

A plot of P-values for the Ljung-Box test at different lags should be produced. If the model fits the data well then these P-values should all be large (i.e. >0.05 at least).

- (b) The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = Z_t + \psi Z_{t-2}$$

where ψ is a real-valued parameter such that $|\psi| < 1$.

- (i) Determine the autocovariance generating function for this process. You should show your working clearly.

SOLUTION:

The process can be written

$$\begin{aligned} Y_t &= (1 + \psi B^2) Z_t \\ &= \alpha(B) Z_t \end{aligned}$$

where $\alpha(z) = 1 + \psi z^2$. The autocovariance generating function is given by

$$\begin{aligned} \Gamma(z) &= \sigma_z^2 \alpha(z) \alpha(z^{-1}) \\ &= \sigma_z^2 (1 + \psi z^2) \left(1 + \frac{\psi}{z^2}\right) \\ &= \sigma_z^2 \left(1 + \psi^2 + \psi z^2 + \frac{\psi}{z^2}\right). \end{aligned}$$

- (ii) Using your answer to (i), write down the autocovariance function for the process $\{Y_t : t \in \mathbb{Z}\}$.

SOLUTION:

The autocovariance at lag h is given by the coefficient of z^h in the autocovariance generating function. Hence the autocovariance function is

$$\gamma_Y(h) = \begin{cases} \sigma_z^2(1 + \psi^2) & h = 0 \\ \sigma_z^2\psi & h = \pm 2 \\ 0 & h = \pm 1, h = \pm 3, \pm 4, \dots \end{cases}$$

CONTINUED

(iii) The process $\{W_t : t \in \mathbb{Z}\}$ is defined

$$W_t = \frac{1}{2} (Y_t + Y_{t-1}).$$

Determine the autocovariance function for this process.

SOLUTION:

Consider for $h \in \mathbb{Z}$:

$$\begin{aligned}\gamma_W(h) &= \text{Cov}(W_t, W_{t+h}) \\ &= \mathbb{E}(W_t W_{t+h}) - \mathbb{E}(W_t) \mathbb{E}(W_{t+h})\end{aligned}$$

Note that $\mathbb{E}(W_t) = 0$ for all t since

$$\begin{aligned}\mathbb{E}(W_t) &= \frac{1}{2} (\mathbb{E}(Y_t) + \mathbb{E}(Y_{t-1})) \\ &= \frac{1}{2} (\psi \mathbb{E}(Z_{t-2}) + \mathbb{E}(Z_t) + \psi \mathbb{E}(Z_{t+h-2}) + \mathbb{E}(Z_{t+h})) \\ &= 0.\end{aligned}$$

Then

$$\begin{aligned}\mathbb{E}(W_t W_{t+h}) &= \frac{1}{4} \mathbb{E}[(Y_t + Y_{t-1})(Y_{t+h} + Y_{t+h-1})] \\ &= \frac{1}{4} [\mathbb{E}(Y_t Y_{t+h}) + \mathbb{E}(Y_{t-1} Y_{t+h}) + \mathbb{E}(Y_t Y_{t+h-1}) + \mathbb{E}(Y_{t-1} Y_{t+h-1})] \\ &= \frac{1}{4} [2\gamma_Y(h) + \gamma_Y(h+1) + \gamma_Y(h-1)]\end{aligned}$$

When $h = 0$:

$$\begin{aligned}\gamma_Y(0) &= \frac{1}{2} \gamma_Y(0) \\ &= \frac{\sigma_z^2}{2} (1 + \psi^2).\end{aligned}$$

When $h = \pm 1$:

$$\begin{aligned}\gamma_Y(1) = \gamma_Y(-1) &= \frac{1}{4} [\gamma_Y(0) + \gamma_Y(2)] \\ &= \frac{\sigma_z^2}{4} [1 + \psi + \psi^2].\end{aligned}$$

When $h = \pm 2$:

$$\begin{aligned}\gamma_Y(2) = \gamma_Y(-2) &= \frac{1}{4} [2\gamma_Y(2)] \\ &= \frac{\sigma_z^2}{2} \psi.\end{aligned}$$

When $h = \pm 3$:

$$\gamma_Y(3) = \gamma_Y(-3) = \frac{1}{4} [\gamma_Y(2)]$$

TURN OVER

$$= \frac{\sigma_z^2}{4}\psi.$$

and $h \geq 4$ or $h \leq -4$, $\gamma_Y(h) = 0$.

Hence

$$\gamma_Y(h) = \begin{cases} \frac{\sigma_z^2}{2}(1 + \psi^2) & h = 0; \\ \frac{\sigma_z^2}{4}(1 + \psi + \psi^2) & h = \pm 1; \\ \frac{\sigma_z^2}{2}\psi & h = \pm 2; \\ \frac{\sigma_z^2}{4}\psi & h = \pm 3; \\ 0 & \text{otherwise.} \end{cases}$$